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Utilizing MOF (UiO-67)-Modified Electrodes for Hydrogen Production from Produced Water: Recycling Industrial Effluents for Green Energy

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Abstract

This study investigates the development of MOF-modified electrodes (UiO-67) as electrocatalysts for the hydrogen evolution reaction (HER) using produced water as a sustainable electrolyte. The MOFs were modified with cobalt and nickel via electrochemical deposition, and the structural and the electrochemical characterizations (PXRD, FTIR, SEM, voltammetry) confirmed successful metal incorporation and enhanced electrocatalyst performance. Ni@UiO-67 showed superior HER activity in brine produced water (BPW), reaching −40 mA/cm². Notably, Co@UiO-67 exhibited the highest performance in real produced water (RPW), reaching ca. −65 mA/cm², reflecting strong impurity tolerance. This work is among the first to systematically assess MOF-based electrocatalysts in both brine and real produced water, highlighting the role of MOF structure and metal choice in enabling scalable and sustainable hydrogen production from industrial wastewater.

Introduction

Real produced water (RPW), a major byproduct of oil and gas production, originates from formation water and injected water used in enhanced recovery.(Dickhout et al., 2017) In mature fields, it can comprise up to 98% of extracted fluids. Its complex composition—including hydrocarbons, salts, metals, and NORM—poses treatment challenges. Treatment typically involves pre-treatment, separation, and polishing, depending on water quality and cost.(Igunnu & Chen, 2014) Reusing treated produced water for hydrogen production offers a sustainable solution for wastewater management and freshwater conservation.(Kusworo, Aryanti, & Utomo, 2018) Hydrogen is increasingly recognized as a clean energy carrier due to its high energy density and zero-emission combustion.(Oliveira, Beswick, & Yan, 2021) While conventional production methods are cost-effective, they emit significant CO_2. In contrast, water electrolysis, especially when powered by renewables, offers a carbon-free alternative.(Ursua, Gandia, & Sanchis, 2011) Despite being energy-intensive, its feasibility improves with advanced electrocatalysts and system integration. Limited freshwater availability and high energy demands remain key barriers to large-scale deployment.

(Wang, Wang, Gong, & Guo, 2014)Electrolysis splits water using direct current in an electrolyzer composed of electrodes, an electrolyte, and a separator. The overall reaction is non-spontaneous and requires external energy and catalytic enhancement. Efficient electrolysis depends on materials with high conductivity, catalytic activity, and durability.(Hassan et al., 2024) Platinum, IrO_2, and RuO_2 are efficient but costly, prompting research into earth-abundant alternatives. Metal–Organic Frameworks (MOFs) have emerged as promising low-cost electrocatalysts due to their high surface area, tunable porosity, and chemical versatility. (Peng, Sanati, Morsali, & García, 2023) Zr-based MOFs like UiO-67 offer structural stability and large pores, making them ideal for catalytic modification.(Figueroa-Quintero et al., 2023) For example, Jiao et al. developed a CoP/reduced graphene oxide (CoP/rGO) nanocomposite from a MOF/GO sandwich structure.(Jiao, Zhou, & Jiang, 2016) The resulting CoP/rGO-400 exhibited excellent HER and OER performance, with overpotentials of 105 mV and 340 mV, respectively, surpassing conventional Pt/C and IrO_2 catalysts. Melillo et al. studied UiO-66-based MOF-on-MOF composites.(Melillo et al., 2023) The UiO-66(Zr)-NH_2@UiO-66(Ce) structure demonstrated effective water splitting, yielding 708 μmol g^{-1} H_2 and 320 μmol g^{-1} O_2 over 22 hours. In another advancement, Yan et al. engineered a MOF–COF hybrid (MOF-808@TpPa-1-COF), achieving a hydrogen evolution rate of 11.88 mmol g^{-1} h^{-1}, 5.6 times higher than the pure COF—due to enhanced charge separation and interface conductivity. (Zhang et al., 2021) Metallic coatings with cobalt (Co) and nickel (Ni) enhance MOF conductivity, introduce active catalytic sites, and improve HER efficiency in both media. Cobalt has shown strong potential in catalyzing HER and OER, with frameworks like ZIF-67, MIL-101, and UiO-type MOFs reporting reduced overpotentials and enhanced current densities upon Co integration.(Chen & Guan, 2024; Dionigi, Reier, Pawolek, Gliech, & Strasser, 2016; Loni, Shokuhfar, & Siadati, 2021; Yu, Budiyanto, & Tüysüz, 2022) Similarly, Ni-loaded MOFs improve charge transfer kinetics and catalytic stability in both acidic and alkaline media. These enhancements are attributed to: (i) the introduction of redox-active sites, (ii) improved conductivity through the formation of metal–carbon composites, and (iii) increased defect density and surface roughness that expose more active sites. A study by Ke Yu et al. demonstrated a Ni-doped Co-MOF-74 integrated with $Ti_3C_2T_x$ MXene (CoNi-MOF-74/MXene/NF), which showed outstanding HER and OER performance in alkaline conditions.(Yu et al., 2022) For OER, it achieved a current density of 100 mA/cm² at a low overpotential of 256 mV, accompanied by a Tafel slope of 40.21 mV/dec. For HER, it reached 10 mA/cm² at an overpotential of 102 mV. Furthermore, when employed in a two-electrode electrolyzer configuration with the same material serving as both anode and cathode, the system required only 1.49 V to deliver a current density of 10 mA/cm², indicating highly efficient overall water splitting performance. Another study has enhanced a cobalt-based MOF through zinc ion doping—either post-synthetically (Co-MOF@Zn) or in situ (Co/Zn-MOF)—followed by pyrolysis at 800 °C. The resulting materials, Co-MOF@Zn-800 and Co/Zn-MOF-800, exhibited lower HER overpotentials (218 mV and 236 mV) and better charge transfer (reduced Tafel slopes).(Qin et al., 2022) Co-MOF@Zn-800 maintained stable current for over 40 hours, demonstrating excellent durability. This study focuses on the development of Zr-based MOF electrocatalysts, i.e. UiO-67 modified with Co and Ni, to enhance the HER in BPW and RPW used as an alternative electrolyte.

Material and Methods

Materials and Reagents

All chemicals used in this study were of high purity and purchased from Sigma-Aldrich. These include zirconium tetrachloride ($ZrCl_4$, 99.9%), biphenyl-4,4′-dicarboxylic acid (BPDC, 97%), 1,3,5-benzenetricarboxylic acid (BTC), N, N-dimethylformamide (DMF, 99.8%), glacial acetic acid (≥99.7%), triethylamine (≥99.5%), absolute methanol, sulfuric acid (1.0 M), and Nafion.

Instruments

The crystalline structures of the synthesized MOFs were analyzed using powder X-ray diffraction (PXRD, Rigaku MiniFlex with Cu-Kα radiation). Surface morphology was observed using a Thermo Scientific Quattro SEM. FTIR spectroscopy (Thermo Scientific™ Nicolet™ iS10) was employed for functional group identification. Electrochemical evaluations were performed using a CHI760E Potentiostat.

MOF Synthesis

A solvothermal method was used. $ZrCl_4$ (0.341 mmol) was dissolved in a 3:1 mixture of DMF and acetic acid. BPDC acid (0.320 mmol) was dissolved in 15 mL of DMF with three drops of triethylamine. The solutions were combined and heated at 120 °C for 12 hours. The solid product was filtered, washed with DMF and methanol, and dried at 80 °C overnight.

Brine Produced Water (BPW) Preparation

BPW was synthetically prepared by dissolving specific salts in distilled water: 53.558 g/kg NaCl, 8.515 g/kg $MgCl_2 \cdot 6H_2O$, 18.817 g/kg $CaCl_2 \cdot 2H_2O$, and 1.351 g/kg Na_2SO_4. Each salt was added gradually with complete dissolution before the next addition. The solution was stirred continuously for 24 hours to ensure homogeneity. This synthetic formulation allows controlled, reproducible testing and eliminates any contamination noise.

Electrode Preparation with Cobalt and Nickel Deposition

For cobalt, 4 mL of pH 7 buffer was mixed with four drops of cobalt solution and sonicated. CV was performed for 40 cycles (scan rate 0.02 V s^{-1}, potential range −0.3 to 1.3 V) to deposit cobalt onto the MOF-modified electrode. For nickel, the same steps were followed using pH 10 buffer. This procedure was applied to both UiO-67, resulting in the formation of Ni@UiO-67 and Co@UiO-67.

Ink Preparation and Electrode Coating

10 mg of UiO-67 was dispersed in either distilled water or a 1:1 ethanol-water mixture with two drops of Nafion. The suspension was sonicated (10–20 minutes) and drop-cast (5.0 µL) onto a glassy carbon electrode (GCE), then dried at 40–50 °C. This procedure was applied to both pristine MOFs and their metal-modified counterparts, namely Ni@UiO-67 and Co@UiO-67.

Electrochemical Methodology

HER measurements were performed in BPW using a three-electrode setup: Ag/AgCl (reference), platinum wire (counter), and MOF-modified GCE (working). LSV was carried out from 0.0 V to −1.5 V at 0.05 V s^{-1}. Cyclic voltammetry (CV) was used to facilitate cobalt and nickel deposition.

Results and Discussion

Crystallographic, Spectroscopic, and Morphological Analysis of UiO-67

Fig. 1a illustrates the powder X-ray diffraction (PXRD) pattern of UiO-67. The pattern within 2θ = 5–20° confirms the high crystallinity of the synthesized UiO-67, with intense peaks at 2θ values of 7.5° and 8.7°. The slight shift in peak positions is attributed to differences in the organic linker length, affecting the crystal lattice parameters and suggesting a more compact structure. The average crystal size, calculated using Scherrer's Equation from the 7.5° peak, is 47.84 nm. Fig. 1b presents the FTIR spectrum of UiO-67. The BPDC linker shows characteristic symmetric and asymmetric C−O stretching vibrations around 1390 cm^{-1} and 1580 cm^{-1}, respectively. A peak at 1660 cm^{-1} corresponds to C=O stretching in the carboxylate groups. Peaks at 665 cm^{-1} and 760 cm^{-1} indicate μ^3–O stretching related to Zr–(OC) interactions. The morphology of UiO-67 was examined using SEM, as shown in Fig. 1c&d. The images reveal larger octahedral particles, ranging from 500 nm to several micrometers, with smooth surfaces attributed to the longer biphenyl-dicarboxylic acid linker.

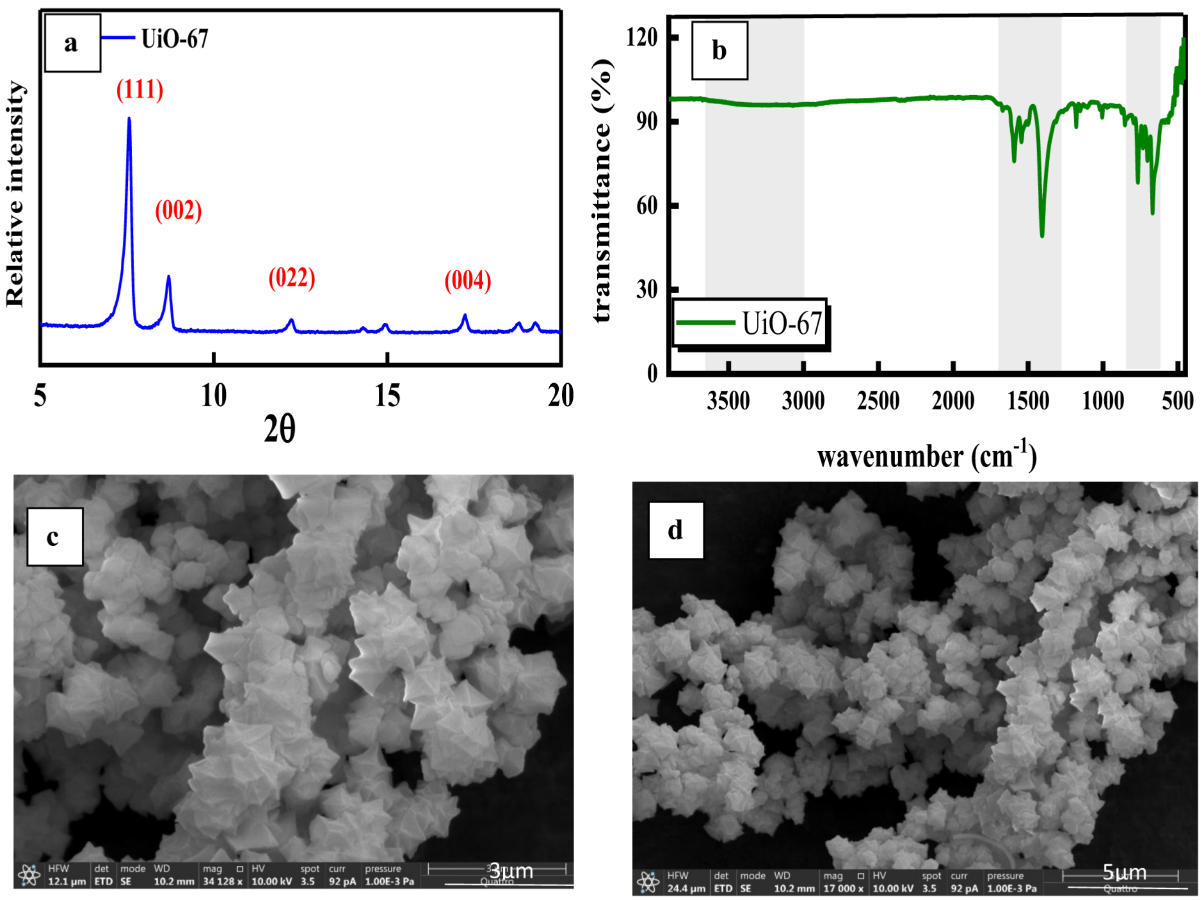

Figure 1—Material characterization: (a) PXRD confirms crystallinity; (b) FTIR indicates carboxylates; (c, d) SEM shows smooth octahedral particles.

HER Activity using UiO-67, Ni@UiO-67, and Co@UiO-67

The electrocatalytic performance of UiO-67-based electrodes was evaluated using Linear Sweep Voltammetry (LSV) in BPW. As shown in Fig. 2a, the pristine UiO-67-modified electrode exhibited enhanced HER activity compared to the bare glassy carbon electrode (GCE), with a cathodic onset potential at ca. −0.6 V and a current density of −1.2 mA/cm², reflecting its porous structure and moderate catalytic capability. Following electrochemical deposition, both Co@UiO-67 and Ni@UiO-67 demonstrated significantly improved HER performance, as illustrated in Fig. 2b. Their current densities reached −25 mA/cm² at more negative potentials (–1.3 V), indicating over a 20-fold increase relative to unmodified UiO-67. Among the two, Ni@UiO-67 showed slightly higher activity, suggesting more favorable HER kinetics. When tested in a mixed electrolyte of produced water and 1 M H_2SO_4, Co@UiO-67 achieved a remarkably high current density (~–230 mA/cm²), surpassing both UiO-67 and cobalt alone. This enhancement, presented in Fig. 2c, is attributed to synergistic interactions between the MOF structure and the deposited cobalt. Further comparisons under various electrolyte conditions confirmed the dominant role of acidity in promoting HER. The addition of 0.4 mL of 1 M H_2SO_4 to BPW yielded the highest current densities for both modified electrodes (~–40 mA/cm²), with Co@UiO-67 slightly outperforming Ni@UiO-67. These findings validate the potential of transition-metal-functionalized UiO-67 as efficient HER catalysts in complex media, e.g. produced water.

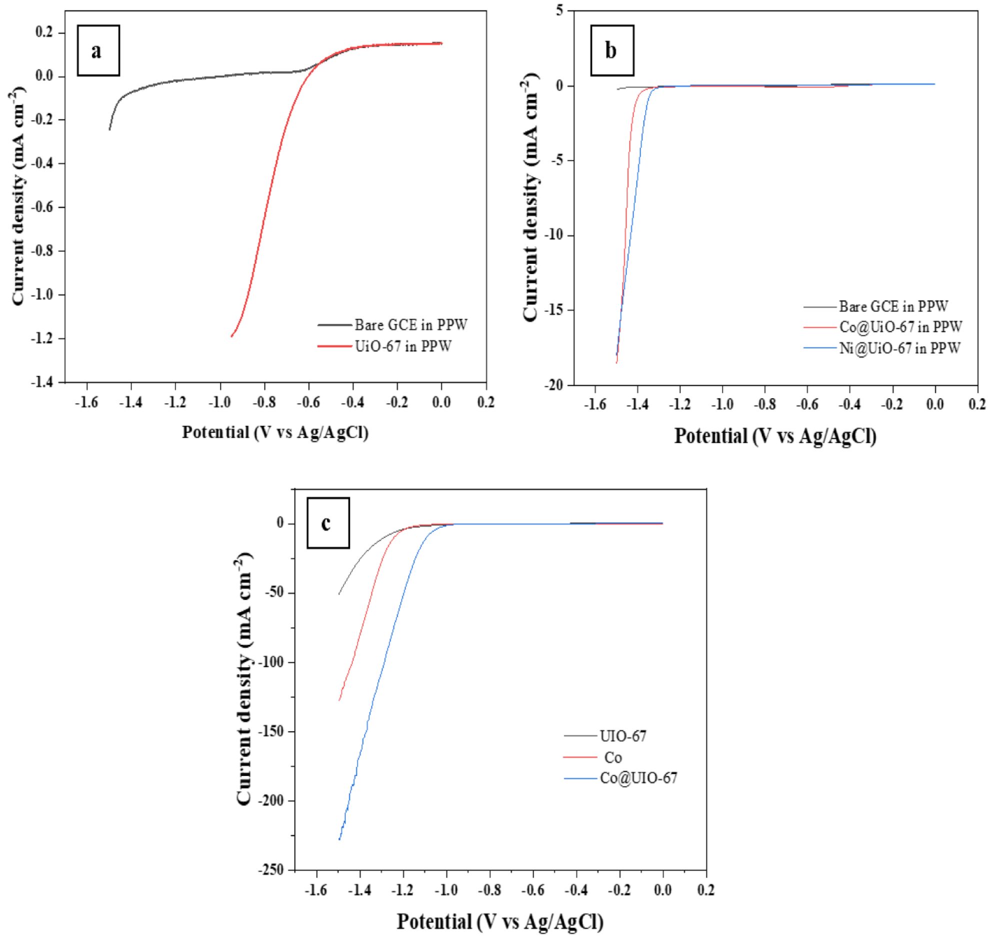

Figure 2—LSV analysis of HER performance for UiO-67-based electrodes: (a) Pristine UiO-67 vs. bare GCE; (b) Co@UiO-67 vs. Ni@UiO-67 in BPW; (c) Enhanced activity of Co@UiO-67 in BPW/1 M H_2SO_4 mixture.

Cyclic Voltammetry (CV) of Ni@UiO-67, and Co@UiO-67

The Cyclic Voltammetry (CV) results presented in Fig. 3a&b highlight the successful electrochemical deposition of Co and Ni onto the UiO-67 framework, forming Co@UiO-67 and Ni@UiO-67 composites. In Fig. 3a, the progressive CV scans for Co@UiO-67 show a clear increase in anodic current beyond 1.0 V vs Ag/AgCl over successive cycles, indicating the onset of Co^{2+} oxidation and accumulation of cobalt species on the MOF surface. This trend confirms efficient and stable cobalt deposition. Similarly, the CV profile for Ni@UiO-67 in Fig. 3b reveals distinct redox features in the potential range of 0.3 V–1.3 V, consistent with the Ni^{2+}/Ni^{3+} redox transitions. The overlapping and increasing CV responses over 20 cycles point to a gradual and controlled buildup of nickel on the MOF, suggesting strong metal–support interaction and stable electrochemical growth. The consistent rise in current response for both systems affirms the feasibility of in-situ metal functionalization of UiO-67 by CV, setting the stage for enhanced electrocatalytic behavior in HER applications.

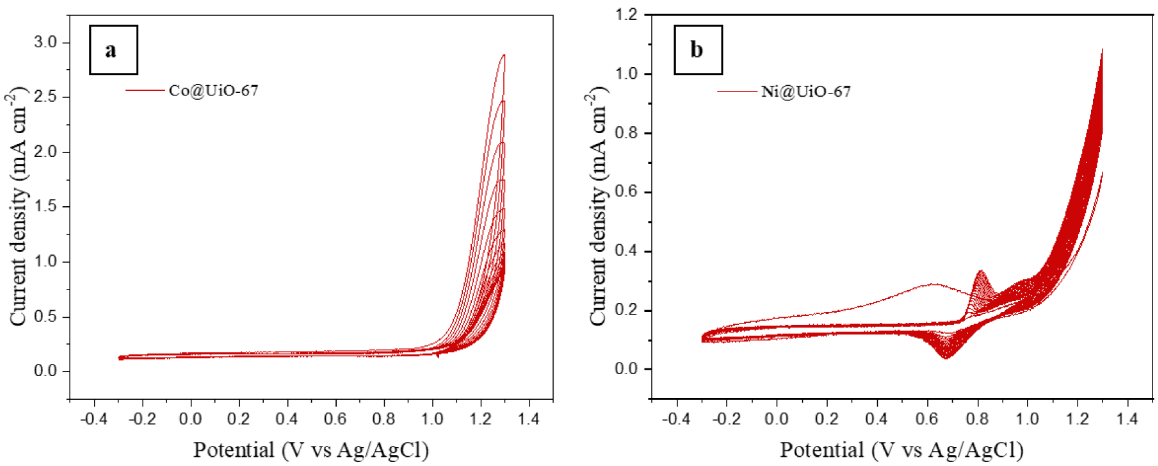

Figure 3—Cyclic voltammograms of Co and Ni deposition on UiO-67: (a) Co@UiO-67; (b) Ni@UiO-67. Scan rate: 0.05 V/s; 20 cycles between 0.3 V and 1.3 V.

Overpotential and Tafel slope of Ni@UiO-67 and Co@UiO-67

The polarization curves in Fig. 4a reveal the overpotentials required to achieve a current density of 10 mA cm^{-2} for Co@UiO-67 and Ni@UiO-67. After converting the potentials to the RHE scale (ca. pH = 0), Ni@UiO-67 exhibits an overpotential of approximately 1060 mV, while Co@UiO-67 requires 1094 mV, indicating better electrocatalytic activity for Ni@UiO-67. The corresponding Tafel plots in Fig. 4b show Tafel slopes of 151.86 mV/dec for Ni@UiO-67 and 161.17 mV/dec for Co@UiO-67. The lower slope for Ni@UiO-67 suggests more favorable HER kinetics and improved charge transfer compared to its cobalt counterpart. Although these values are higher than those of benchmark catalysts like platinum, which typically display slopes around 30 mV/dec and require <50 mV overpotential, the ability of Co@UiO-67 and Ni@UiO-67 to perform HER in a complex, real-world produced water highlights their potential as viable alternatives for hydrogen production in challenging environments.

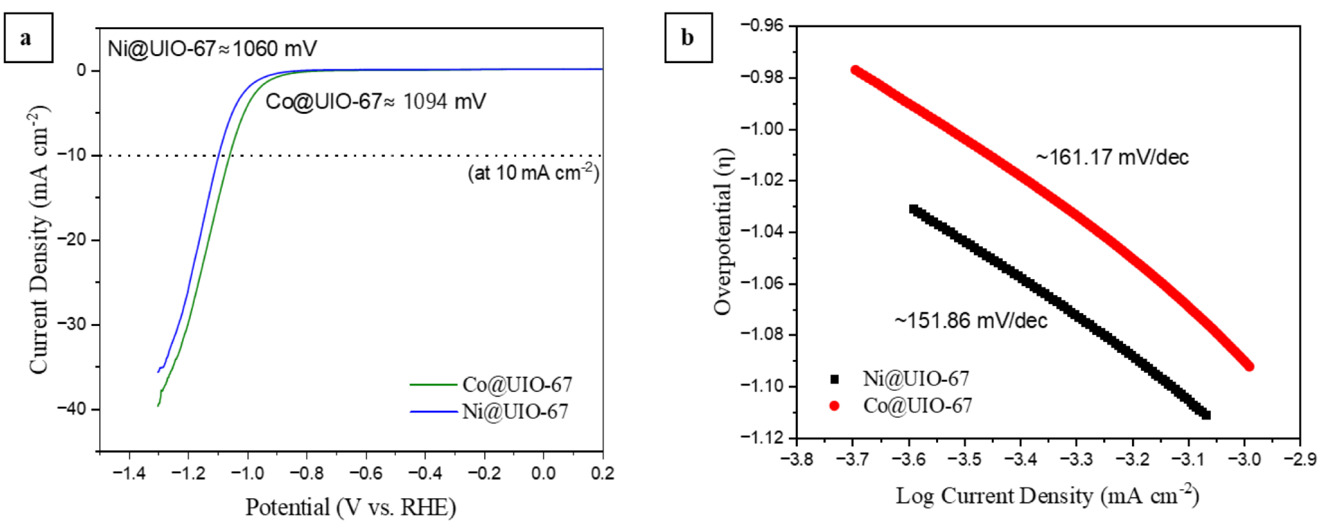

Figure 4—(a) Polarization curves and (b) corresponding Tafel plots of Co@UiO-67 and Ni@UiO-67.

Real Produced Water (RPW) Tests

The electrocatalytic performance of the synthesized MOF-based electrodes was evaluated in RPW acidified with H_2SO_4 to simulate practical application conditions. As shown in Fig. 5a, Co@UiO-67 outperformed Ni@UiO-67, achieving a higher current density of approximately −60 mA cm^{-2} at −1.3 V vs Ag/AgCl, compared to −43 mA cm^{-2} for Ni@UiO-67. Co@UiO-67 also exhibited an earlier onset potential, indicating enhanced HER kinetics in complex water matrices. This performance contrasts with results in synthetic electrolytes, where Ni-based catalysts typically perform better, suggesting that Co-based catalysts may offer greater impurity tolerance in real environments. These findings collectively highlight the superior performance and practical viability of Co-based MOF catalysts in untreated, impurity-rich produced water, emphasizing the importance of real-world testing in catalyst development for sustainable hydrogen production.

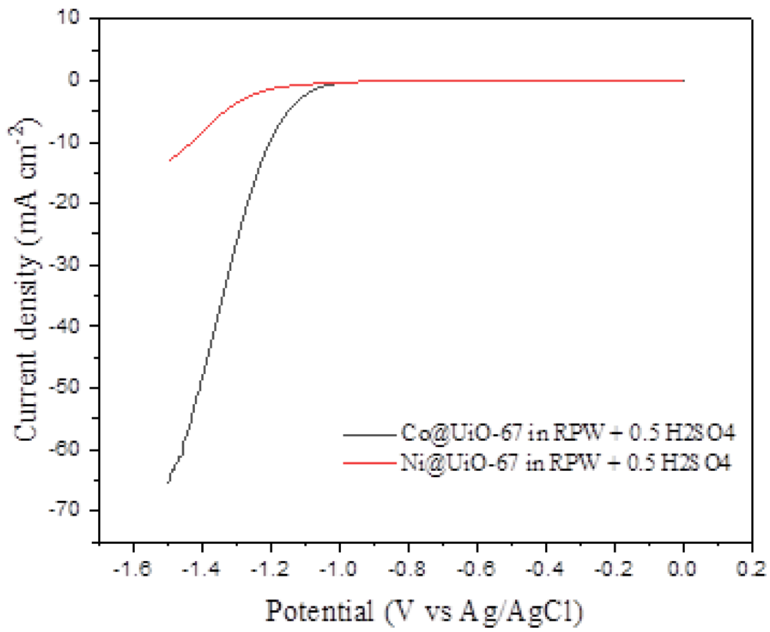

Figure 5—LSV curves of Co@UiO-67 vs. Ni@UiO-67 in H_2SO_4-acidified RPW, showing superior HER performance of Co-based electrodes with higher current density and earlier onset potential.

Conclusion

The electrochemical evaluation of UiO-67 demonstrated its potential as a hydrogen evolution reaction (HER) catalyst in BPW, showing noticeable improvement over the bare glassy carbon electrode (GCE) due to its high surface area and porous architecture. However, the intrinsic activity of UiO-67 remained modest, indicating the need for further enhancement. Electrochemical deposition of cobalt and nickel onto UiO-67 significantly boosted HER performance, with both Co@UiO-67 and Ni@UiO-67 exhibiting early onset potentials and over 20-fold increases in current density compared to pristine UiO-67. Among them, Ni@UiO-67 showed slightly higher catalytic activity under neutral conditions, while Co@UiO-67 demonstrated superior performance in acidic media, reaching to current densities ca. −40 mA/cm². This enhancement is attributed to the synergistic combination of the MOF's porous framework and the catalytic properties of the incorporated metals. The comparative evaluation of UiO-67 under both the brine and the real produced water conditions reveals distinct performance trends influenced by MOF structure, metal modification, and electrolyte composition. In BPW, Ni@UiO-67 exhibited superior performance, achieving current densities up to ca. −40 mA/cm². Conversely, in RPW, Co@UiO-67 demonstrated greater tolerance and electrocatalytic stability, reaching ca. −65 mA/cm², significantly outperforming all other catalysts. These results indicate that while nickel modification is more effective in clean, synthetic environments, UiO-67 particularly when functionalized with cobalt offers better resilience and performance in complex, untreated wastewater. These findings collectively support the development of economical, scalable, and environmentally sustainable hydrogen technologies through the reutilization of industrial wastewater.

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